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# The formation mechanism of TiO<sub>2</sub> polymorphs under hydrothermal conditions based on the structural evolution of [Ti(OH)<sub>h</sub>(H<sub>2</sub>O)<sub>6-h</sub>]<sup>4-h</sup> monomers†

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In the hydrothermal synthesis of TiO<sub>2</sub>, [Ti(OH)<sub>h</sub>(H<sub>2</sub>O)<sub>6-h</sub>]<sup>4-h</sup> (*h* is the hydrolysis ratio) monomers are generated by the dissolution of the precursor containing titanium ions and then, the monomers form TiO<sub>2</sub> crystals *via* a condensation reaction. In this study, the formation mechanism was investigated by the hydrolysis ratio rather than reaction parameters, such as pH value and temperature, and it was proposed that the hydrolysis ratio determines the crystal structure. According to the principle that the oxolation reaction occurs prior to the oxolation reaction, the detailed nucleation process of TiO<sub>2</sub> was derived by a [Ti(OH)<sub>h</sub>(H<sub>2</sub>O)<sub>6-h</sub>]<sup>4-h</sup> monomer with different hydrolysis ratios, and the corresponding relationship between the hydrolysis ratio and the crystal structure was investigated. When the hydrolysis ratio is in the intervals of  $h \leq 2$ ,  $2 < h < 3$ ,  $3 \leq h < 5$  and  $h \geq 5$ , the predicted crystal structures of TiO<sub>2</sub> are rutile, a mixed crystal (rutile, brookite and anatase), anatase and brookite, respectively. Under hydrothermal conditions, the change in temperature leads to the change in the ionization of water, which leads to the deviation in the hydrolysis ratio. In order to solve this problem, the calculation method of hydrolysis ratio (*h*) under different hydrothermal conditions was established by introducing the ionic product correction. Moreover, the hydrolysis ratio dynamically changed by the condensation reaction of a [Ti(OH)<sub>h</sub>(H<sub>2</sub>O)<sub>6-h</sub>]<sup>4-h</sup> monomer during hydrothermal crystallization, and it may shift to the adjacent crystal formation interval, resulting in the formation of the corresponding TiO<sub>2</sub> at a new interval. The crystal structure of TiO<sub>2</sub> synthesized at different hydrolysis ratios by the hydrothermal method with Ti(OH)<sub>4</sub> as the precursor was consistent with the theoretical predictions.

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## Introduction

Titanium dioxide is a compound with many polymorphs, and rutile (R), anatase (A) and brookite (B) are the most common crystal structures.<sup>1,2</sup> These three crystal structures can be regarded as the arrangements of the [TiO<sub>6</sub>] octahedral in different edge/corner sharing connection manners. In recent years, many researchers have successfully synthesized different phases of TiO<sub>2</sub> under hydrothermal conditions.<sup>3–7</sup> The effects of precursors, concentration, hydrothermal temperature and pH value on the synthesized TiO<sub>2</sub> phases have also been studied.<sup>8–12</sup> Currently, the crystallization process of TiO<sub>2</sub> in a hydrothermal synthesis is considered to be the “dissolution-crystallization mechanism”.<sup>13,14</sup> At a certain hydrothermal

temperature, the hydrate containing titanium ions begins to dissolve, forming Ti<sup>4+</sup>-hydrated octahedral monomers. Then, the monomers form polymers *via* a condensation reaction, and result in the formation of TiO<sub>2</sub> nucleus. The connection manner of [TiO<sub>6</sub>] octahedral in the TiO<sub>2</sub> nucleus corresponds to the connection manner of the octahedral monomers. Therefore, the structure of the Ti<sup>4+</sup>-hydrated octahedral monomers and the nucleation process under hydrothermal conditions are the key factors to explore the formation mechanism of TiO<sub>2</sub> polymorphs.

In early 1992, Henry *et al.*<sup>15</sup> studied the hydrolysis of Ti<sup>4+</sup> ions in aqueous solutions and proposed that the structure of a Ti<sup>4+</sup>-hydrated octahedral monomer is [Ti(OH)<sub>h</sub>(H<sub>2</sub>O)<sub>6-h</sub>]<sup>4-h</sup>. The partial charge model was proposed to calculate the relationship between the pH value and the hydrolysis ratio (*h*) at room temperature. However, the partial charge model does not take into account that the hydrolysis ratio changes under various hydrothermal conditions. Yanqing *et al.*<sup>16–18</sup> studied the influence of the concentration and pH value on the hydrothermal synthesis of TiO<sub>2</sub> crystallites using TiCl<sub>4</sub> as the precursor. With a significance on the regulation of TiO<sub>2</sub> crystals

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under hydrothermal conditions, the nucleation processes of rutile, anatase and brookite crystals were proposed. However, they did not consider the necessary relation between the hydrolysis ratio and crystal structure, and the quantitative rule of the formation of  $\text{TiO}_2$  crystal was also not included. Moreover, it is not appropriate to explain the nucleation process and the crystal transformation using the pH value, temperature and other reaction parameters. At present, the relationship between the pH value and the structure of a  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer under hydrothermal conditions is not clear, and the relationship between the hydrolysis ratio and crystal structure needs further explanations. In addition, the fact that the hydrolysis ratio may dynamically change during hydrothermal synthesis has not given due consideration, and the way to generate a mixed crystal may also be different. Therefore, these problems make it difficult to achieve the precise regulation of the crystal structure of  $\text{TiO}_2$ .

The purpose of this study is to further study the nucleation process of  $\text{TiO}_2$  under hydrothermal conditions. Based on the difference in the  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer structure, leading to the difference in edge/corner-shared connection manners, a new crystal formation mechanism is proposed. The nucleation process of  $\text{TiO}_2$  is derived by a  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer with different hydrolysis ratios, and the relationship between the hydrolysis ratio and the crystal structure of  $\text{TiO}_2$  is clearly defined. Then, the relationship between the hydrolysis ratio and pH value under hydrothermal conditions is concluded by introducing the ionic product of water for correction. Thus,  $\text{TiO}_2$  is synthesized by the hydrothermal method at different hydrolysis ratios with  $\text{Ti}(\text{OH})_4$  as a precursor. The crystal phase of the obtained sample was determined using XRD to verify the proposed formation mechanism of  $\text{TiO}_2$  with different polymorphs.

## Results and discussions

### Crystal formation mechanism of $\text{TiO}_2$

The local order in rutile, anatase and brookite  $\text{TiO}_2$  is represented by the arrangements of the  $[\text{TiO}_6]$  octahedra, as shown in Fig. 1.<sup>19,20</sup> The  $[\text{TiO}_6]$  octahedron in the rutile structure is surrounded by ten octahedra, in which two are edge-shared and eight are corner-shared. The  $[\text{TiO}_6]$  octahedron in the anatase structure is surrounded by eight octahedra, in which four are edge-shared and four are corner-shared. The  $[\text{TiO}_6]$  octahedron in the brookite structure is surrounded by nine octahedra, in which three are edge-shared and six are corner-shared.

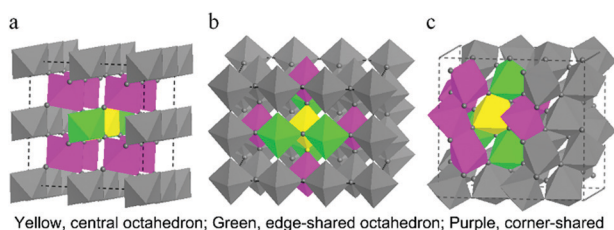
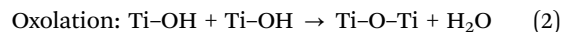
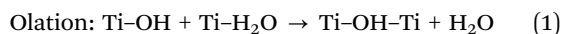


Fig. 1 The arrangement of  $[\text{TiO}_6]$  octahedron in  $\text{TiO}_2$  polymorphs. (a) Rutile; (b) anatase; (c) brookite.

Therefore, the crystal structure of  $\text{TiO}_2$  can be summarized by the different types of edge-shared and corner-shared  $[\text{TiO}_6]$  octahedra.

The type of edge-shared and corner-shared octahedra is related to the structure of the octahedral  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer. The octahedral monomer contains two kinds of bonds, which are  $h$  Ti–OH bonds and  $(6 - h)$  Ti– $\text{H}_2\text{O}$  bonds. The reaction between the two bonds removes a water molecule and the octahedra connect with each other *via* corner/edge sharing. The condensation of octahedral monomers follows two main reactions: ololation and oxolation.<sup>21</sup>



The activation energy of the ololation and the oxolation reaction is  $20 \text{ kJ mol}^{-1}$  and  $60 \text{ kJ mol}^{-1}$ , respectively. Thus, the ololation reaction will occur prior to the oxolation reaction. The difference in the number and position of Ti–OH bonds and Ti– $\text{H}_2\text{O}$  bonds in the octahedral monomer with different hydrolysis ratios leads to the difference in the ololation or the oxolation reaction. As a result, the edge/corner sharing connection manners between the monomers are different, and  $\text{TiO}_2$  with different crystal structures is formed. Here, we have proposed a new formation mechanism of  $\text{TiO}_2$  with different polymorphs at the molecular level, and the nucleation process of  $\text{TiO}_2$  was derived when the hydrolysis ratio was between 2 and 5. We begun with the hydrolysis ratio of 4, when the  $[\text{Ti}(\text{OH})_4(\text{H}_2\text{O})_2]$  monomers showed neutrality and high symmetry.

When  $h = 4$ , two octahedral  $[\text{Ti}(\text{OH})_4(\text{H}_2\text{O})_2]$  monomers give priority to form an edge-sharing dimer *via* the ololation reaction. Then, the trimers and the pentamers with higher symmetry were formed *via* the ololation reaction by edge sharing. The pentamers continue to form the crystal nucleus of anatase in this way, and the nucleation process is shown in Fig. 2.

When  $h = 3$ , the nuclei formation of octahedral  $[\text{Ti}(\text{OH})_3(\text{H}_2\text{O})_3]^+$  monomers is similar to that when  $h = 4$ , and the nucleation process is shown in Fig. 2. Compared with  $h = 4$ , the amount of  $\text{H}_2\text{O}$  (blue) is high in the polymer structures.

When  $h = 2$ , octahedral  $[\text{Ti}(\text{OH})_2(\text{H}_2\text{O})_4]^{2+}$  monomers give priority to form an edge-sharing straight chains *via* the ololation reaction. However, it needs to be pointed out that the straight chain is formed in a gradient direction rather than in a horizontal direction. The straight chains are further connected by corner sharing, finally forming the crystal nucleus of the rutile, and the nucleation process is shown in Fig. 3.

When  $2 < h < 3$ , the crystal nucleus of rutile and anatase is formed by the self-polymerization of  $[\text{Ti}(\text{OH})_2(\text{H}_2\text{O})_4]^{2+}$  and  $[\text{Ti}(\text{OH})_3(\text{H}_2\text{O})_3]^+$  monomers, respectively. At the same time, three  $[\text{Ti}(\text{OH})_2(\text{H}_2\text{O})_4]^{2+}$  monomers form an edge-sharing trimer *via* the ololation reaction, and then the tetramer is formed by connecting with another  $[\text{Ti}(\text{OH})_3(\text{H}_2\text{O})_3]^+$  monomer. The octahedron at the center is connected to three other octahedra in the tetramer, and it is continuously polymerized into a polymer with a double layer in this manner. The crystal nucleus of brookite is formed *via* the connection of double layers by corner

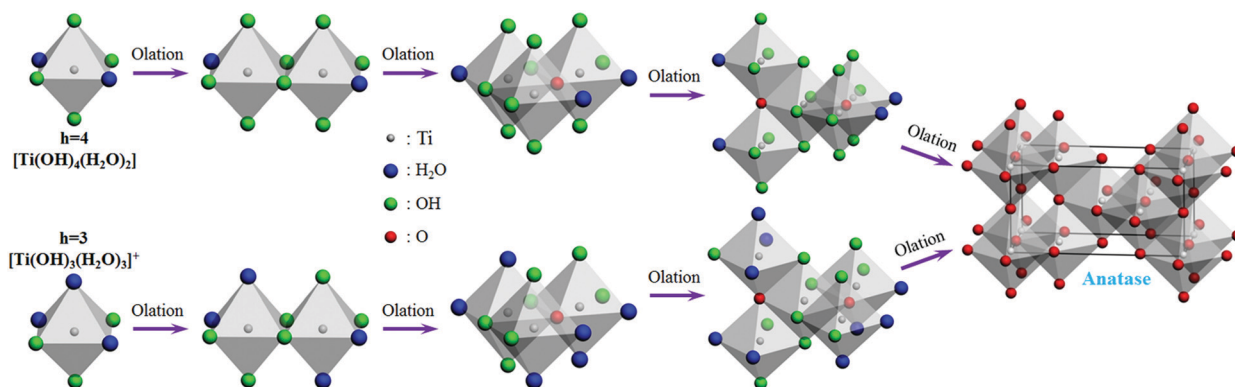


Fig. 2 A possible condensation pathway for the nucleation of anatase at  $h = 3$  and 4.

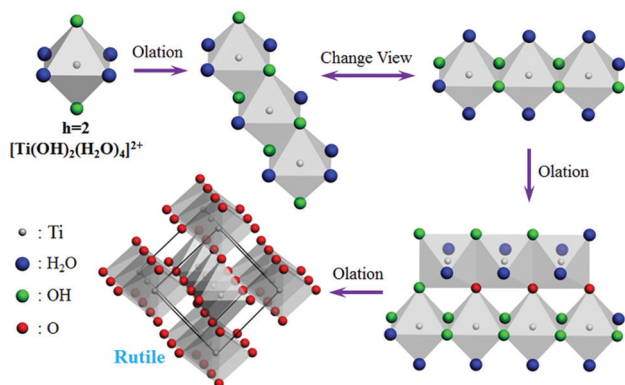


Fig. 3 A possible condensation pathway for the nucleation of rutile at  $h = 2$ .

sharing, and the nucleation process is shown in Fig. 4. Therefore, when  $2 < h < 3$ , a mixed crystal of rutile, brookite and anatase is obtained.

When  $h = 5$ , two octahedral  $[\text{Ti}(\text{OH})_5(\text{H}_2\text{O})]^-$  monomers form an edge-sharing dimer *via* the olation reaction, and the tetramer is formed. Then, the eight polymer and sixteen polymer are gradually formed *via* the oxolation reaction by edge sharing. Finally, the crystal nucleus of brookite is formed *via* the connection of double layers of the sixteen polymer by

corner sharing, and the nucleation process is shown in Fig. 4. As the activation energy of the olation reaction is much higher than that of the oxolation reaction, the reaction temperature required for this nucleation process is higher than that of other nucleation processes.

The calculation of the hydrolysis ratio of hydrated titanium ions at room temperature is given by eqn (3).<sup>18</sup>

$$h = (1.0584 + 0.3181 \text{ pH}) / (0.6794 + 0.01776 \text{ pH}) \quad (3)$$

From eqn (3), when  $h = 1$  and 6, the corresponding pH value was  $-1.2$  and  $14.5$ , respectively. Under these two conditions,  $\text{TiO}_2$  was difficult to crystallize by the hydrothermal method, so we did not deduce the nucleation process at  $h = 1$  and 6.

The crystal structure at different hydrolysis ratios and the regularity of crystal transformation are summarized in Fig. 5. It can be concluded that  $h = 2, 3$  and 5 are the key points of crystal transformation, and  $h \leq 2$ ,  $2 < h < 3$ ,  $3 \leq h < 5$  and  $h \geq 5$  are the four crystal formation intervals. When  $h \leq 2$ , the structure is crystallized in the rutile structure. When  $2 < h < 3$ , a mixed crystal of rutile, brookite and anatase will appear. In this case, when  $h$  is close to 2, the rutile structure is seen in a higher proportion, whereas when  $h$  is close to 3, the anatase structure is seen in a higher proportion. When  $3 \leq h < 5$ , only the

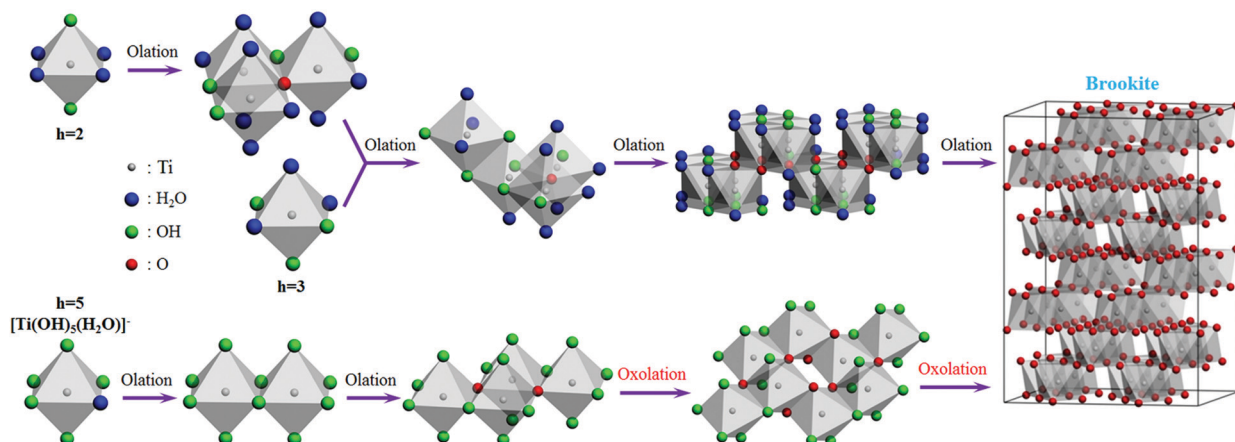


Fig. 4 A possible condensation pathway for the nucleation of brookite at  $2 < h < 3$  and  $h = 5$ .

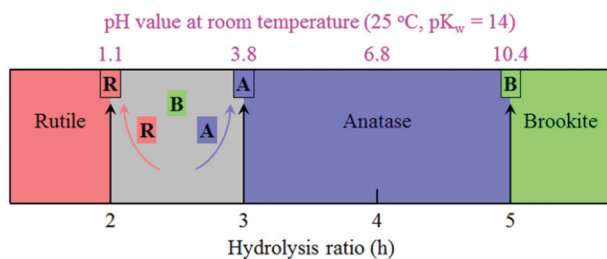
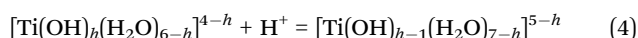


Fig. 5 The relationship between the crystal structure and the hydrolysis ratio. The corresponding pH values at room temperature (25 °C,  $pK_w = 14$ ) were also labeled.

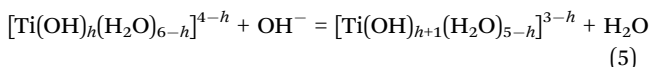
anatase structure will crystallize. Finally, with higher  $h$  ( $h \geq 5$ ), only the brookite structure appears.

In addition, under the hydrothermal conditions, the ionic product constant ( $K_w$ ) of water increases, and the concentration of  $H^+$  and  $OH^-$  changes at various temperatures. Therefore, the hydrolysis ratio at a high temperature is not exactly the same as that at room temperature.

If the concentration of  $H^+$  is increased, the hydrolysis ratio will decrease, as shown in eqn (4):



If the concentration of  $H^+$  is decreased, the hydrolysis ratio will increase, as shown in eqn (5):



Under strongly acidic conditions,  $H^+$  is supplied by a strong acid, and  $OH^-$  is generated by the ionization of water. When  $K_w$  increases the concentration of  $OH^-$  increases, resulting in an increase in the hydrolysis ratio. Under strongly basic conditions,  $OH^-$  is supplied by a strong base, and  $H^+$  is generated by the ionization of water. When  $K_w$  increases the concentration of  $H^+$  increases, resulting in a decrease in the hydrolysis ratio. Under neutral conditions, when  $K_w$  increases, the concentrations of  $H^+$  and  $OH^-$  vary slightly, and the hydrolysis ratio does not change significantly.

In order to verify the crystal nucleation mechanism and the relationship between the hydrolysis ratio and the pH value, further experimental studies were performed. The pH value of the purified  $Ti(OH)_4$  suspension was adjusted by HCl or NaOH to 1.1, 6.8 and 10.4. According to eqn (3), the hydrolysis ratio at room temperature was 2, 4 and 5. Then, the suspension was hydrothermally treated at 150 °C for 18 h, and the crystal structures of the as-prepared samples are shown in Fig. 6. At pH = 1.1, the as-prepared  $TiO_2$  is not purely rutile but a mixed crystal structure of rutile, brookite and anatase. This result is not consistent with the previous theoretically predicted result of  $h = 2$ , but similar to the result of the higher hydrolysis ratio. Pure anatase was obtained at pH = 6.8, which is in agreement with the theoretical result of  $h = 4$ . At pH = 10.4, the as-prepared  $TiO_2$  is not purely brookite but a mixed crystal structure of brookite and anatase. This result is not consistent with the previous theoretically predicted result of  $h = 5$ , but similar to the result of the lower hydrolysis ratio. There is a certain

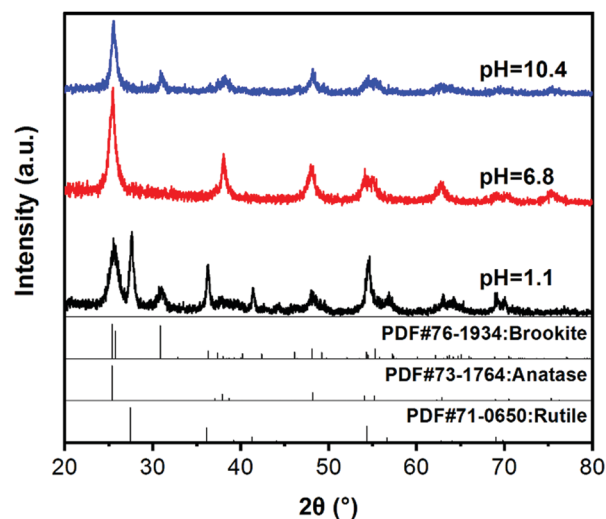


Fig. 6 XRD patterns of the samples synthesized at different pH values.

deviation between the experimental results at different pH values and the theoretical results, corresponding to the hydrolysis ratio at room temperature.

We have also consulted the relevant results in the literature, as shown in Table 1. Comparing the results of the crystal structures of  $TiO_2$  synthesized by the hydrothermal method under different pH values to our theoretical results, we found the same problem. The experimental results reported in literature are in accordance with our theoretical results at pH = 3–9, and there are also deviations at pH = 1 and pH = 11. In order to accurately regulate the crystal phase, the hydrolysis ratio must be modified to obtain the quantitative relationship between the hydrolysis ratio and the pH value under hydrothermal conditions.

### Hydrolysis ratio under hydrothermal conditions

We calculated the hydrolysis ratio of hydrated titanium ions based on the partial charge model proposed by Livage and Henry.<sup>15</sup> The basic principle of this model is that when two or more atoms with different electronegativity are combined, the electron transfer depends on the difference of electronegativity between them. When the electronegativity is equal, the electron transfer stops.

The mean electronegativity ( $\chi$ ) of the hydrolysates is given by eqn (6)

$$\chi = \frac{\sum_i \sqrt{\chi_i^0} + 1.36Z}{\sum_i 1/\sqrt{\chi_i^0}} \quad (6)$$

Table 1 The pH value and crystal structure in some literature

| pH value         | $h$ at room temperature | Theoretical phase | Experimental phase |
|------------------|-------------------------|-------------------|--------------------|
| 1 <sup>22</sup>  | 1.9                     | R                 | B, A, R            |
| 3 <sup>23</sup>  | 2.7                     | A, B              | A, B               |
| 5 <sup>23</sup>  | 3.4                     | A                 | A                  |
| 7 <sup>23</sup>  | 4.1                     | A                 | A                  |
| 9 <sup>23</sup>  | 4.6                     | A                 | A                  |
| 11 <sup>23</sup> | 5.2                     | B                 | A, B               |

where  $\chi_i^0$  is the electronegativity of each neutral atom in the chemical species. The electronegativity of Ti, O and H atoms are 1.32, 3.5 and 2.1, respectively.<sup>24</sup>  $Z$  is the total charge and equals to the sum of the partial charges ( $\delta_i$ ) of all individual atoms:  $Z = \sum \delta_i$ . The partial charges ( $\delta_i$ ) are given by eqn (7)

$$\delta_i = \frac{\chi - \chi_i^0}{1.36\sqrt{\chi_i^0}} \quad (7)$$

For hydrated titanium ions  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$ , eqn (8) can be obtained by charge conservation.

$$\delta(\chi, \text{Ti}) + 6\delta(\chi, \text{O}) + (12 - h)\delta(\chi, \text{H}) = 4 - h \quad (8)$$

For aquo-ions  $[\text{Ti}(\text{H}_2\text{O})_6]^{4+}$ , the summation of partial charges is given by eqn (9)

$$\delta[\chi, \text{Ti}(\text{H}_2\text{O})_6] = \delta(\chi, \text{Ti}) + 6\delta(\chi, \text{O}) + 12\delta(\chi, \text{H}) \quad (9)$$

According to eqn (8) and (9), the hydrolysis ratio is given by eqn (10)

$$h = \frac{4 - \delta[\chi, \text{Ti}(\text{H}_2\text{O})_6]}{1 - \delta(\chi, \text{H})} \quad (10)$$

The mean electronegativity ( $\chi_h$ ) of the  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  is equal to the mean electronegativity ( $\chi_w$ ) of the aqueous solution.

$$\chi_h = \chi_w = 2.7321 - 0.0345 \text{ pH} \quad (11)$$

$\delta[\chi, \text{Ti}(\text{H}_2\text{O})_6]$  and  $\delta(\chi, \text{H})$  can be calculated from eqn (11) and (7), and then the hydrolysis ratio can be calculated from eqn (10).

$$h = (1.0584 + 0.3131 \text{ pH}) / (0.6793 + 0.01749 \text{ pH}) \quad (12)$$

Eqn (12) is the relationship between the hydrolysis ratio and the pH value at room temperature, which is in accordance with eqn (3) reported in the literature. There is no parameter related to temperature in eqn (12), so it cannot express the possible change in the hydrolysis ratio under different temperatures. Since neutral water is always neutral under the different ionic products ( $K_w$ ) and the hydrolysis ratio does not change, we have proposed a correction method based on the ionic product of water. Eqn (13) is obtained using the mathematical transformation of eqn (11)

$$\chi_w = 2.7321 - 0.0345 \text{ pH} = 2.4908 + 0.0345 \times (14/2 - \text{pH}) \quad (13)$$

At room temperature,  $K_w = 1 \times 10^{-14}$ ; therefore eqn (14) can be derived as

$$\chi_h = \chi_w = 2.4908 + 0.0345 \times (\text{p}K_w/2 - \text{pH}) \quad (14)$$

Using eqn (14) instead of eqn (11), the hydrolysis ratio can be calculated by eqn (15).

$$h = \frac{3.2507 - 0.1566 + \text{p}K_w + 0.3131\text{pH}}{0.80168 - 0.00874 + \text{p}K_w + 0.01749\text{pH}} \quad (15)$$

It must be pointed out that the pH value in eqn (15) is under hydrothermal conditions (varied  $\text{p}K_w$ ), which is not always the same as that under ambient conditions ( $\text{p}K_w = 14$ ). The relationships

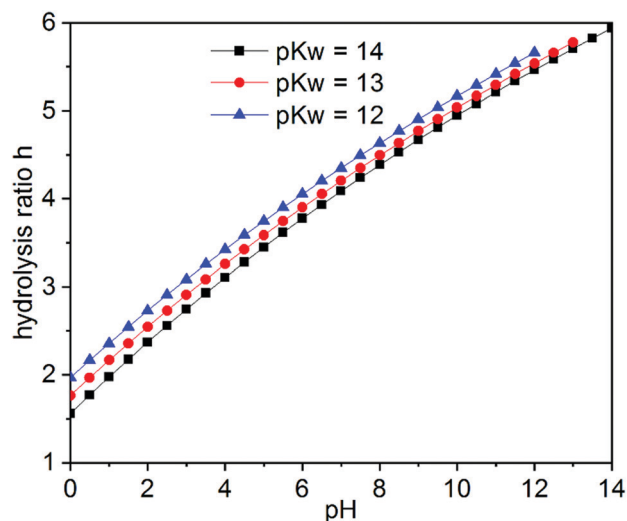


Fig. 7 The relationship between the hydrolysis ratio and the pH value.

of hydrolysis ratio and its corresponding pH value under different ionic products ( $K_w$ ) are shown in Fig. 7. For a solution with  $\text{pH} = 0$ , when  $\text{p}K_w$  decreases from 14 to 12,  $h$  increases from 1.56 to 1.97. For a solution with  $\text{pH} = 7$ , when  $\text{p}K_w$  decreases from 14 to 12,  $\text{pH}$  value also decreases to 6 and  $h$  remains at 4.05. For a solution with  $\text{pH} = 14$ , when  $\text{p}K_w$  decreases from 14 to 12, the  $\text{pH}$  value also decreases to 12 and  $h$  decreases from 5.89 to 5.66. This is in agreement with the qualitative analysis of the change in the hydrolysis ratio under hydrothermal conditions. Therefore, it is reasonable to introduce the ionic product ( $K_w$ ) for the correction of the relationship between the hydrolysis ratio and the pH value.

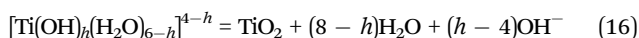
In the process of the hydrothermal synthesis of  $\text{TiO}_2$ , the ionic product ( $K_w$ ) of water is difficult to be directly obtained by measurement. The ionic product is mainly related to the temperature and the ionic strength, so we have consulted the ionic product under different temperatures and ionic strengths. As  $h = 2, 3$  and  $5$  are the key points of crystal transformation, the  $\text{pH}$  values at room temperature are calculated for the crystal transformation points under different hydrothermal conditions, as shown in Table S1 (ESI†). Combining the relationship between the pH value and the hydrolysis ratio (Table S1, ESI†) with the relationship between the hydrolysis ratio and the crystal structure (Fig. 5), the crystal structure of  $\text{TiO}_2$  prepared by the hydrothermal synthesis can be easily predicted.

According to the relationship between the hydrolysis ratio and pH value corrected by the ionic product, we further analysed the experimental results in Fig. 6. When  $\text{pH} = 1.1$ ,  $\text{p}K_w = 11.2$ , and  $h = 2.5$ , the prepared  $\text{TiO}_2$  is the three-phase mixed crystal instead of pure rutile, which is in accordance with the experimental result of  $\text{pH} = 1.1$ . When  $\text{pH} = 10.4$ ,  $\text{p}K_w = 11.3$ , and  $h = 4.6$ , the prepared  $\text{TiO}_2$  is anatase instead of brookite, which is also consistent with the experimental result of  $\text{pH} = 10.4$ . However, at  $\text{pH} = 10.4$ , there is a small amount of brookite due to the dynamic change in the hydrolysis ratio during the hydrothermal process and has been discussed in the next section.

### Dynamic change of hydrolysis ratio

In the process of hydrothermal synthesis of  $\text{TiO}_2$ , the pH value of the solution may change, so the hydrolysis ratio will dynamically change with the pH value. Particularly, when the initial hydrolysis ratio is near the crystal transformation point, the hydrolysis ratio may shift to the adjacent crystal formation interval, resulting in the formation of a new crystal structure of  $\text{TiO}_2$ . We will further study these changes.

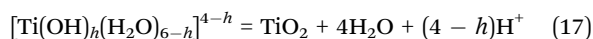
In order to study the crystal changes around initial  $h = 5$ , the pH value of  $\text{Ti}(\text{OH})_4$  suspension is adjusted by NaOH, and the temperature of hydrothermal crystallization is 125 °C or 150 °C with the holding time of 18 h, and the results are shown in Fig. 8a. When the pH value is 11.6 and the hydrothermal temperature is 125 °C, amorphous  $\text{TiO}_2$  is obtained. Under these hydrothermal conditions, the  $pK_w$  is about 11.6, and the corresponding hydrolysis ratio is 5 according to Table S1 (ESI†). This shows that the activation energy of crystallization of brookite is higher at  $h = 5$ , which is consistent with the need for the oxolation reaction during the nucleation process. When the pH value is 10.4 and the hydrothermal temperature is 150 °C, the prepared  $\text{TiO}_2$  is mainly anatase with a small amount of brookite. With the increase in the pH value, the content of brookite increases and the content of anatase decreases correspondingly. At pH = 11.7, the prepared  $\text{TiO}_2$  is pure brookite, which is in agreement with the theoretical result of  $h = 5$ . At  $h < 5$ , the prepared  $\text{TiO}_2$  contains a small amount of brookite because of the dynamic change in the hydrolysis ratio during the hydrothermal process. At  $h > 4$ ,  $\text{OH}^-$  is constantly generated by the condensation reaction of a  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer during the crystallization process, resulting in a continuous increase in the pH value, as shown in eqn (16).



The pH value increased from the initial 10.4 to the final 12.5, so the hydrolysis ratio increased from 4.6 to 5.3 during the hydrothermal process. As brookite can be obtained at  $h \geq 5$ , there is a small amount of brookite at pH = 10.4.

In order to study the crystal changes around the initial  $h = 3$  and 2, the  $\text{Ti}(\text{OH})_4$  suspension adjusted to different pH values by adding HCl was hydrothermally treated at 125 °C for 12 h, and the results are shown in Fig. 8b and c.

At pH = 1.5, the prepared  $\text{TiO}_2$  is mainly brookite with a small amount of rutile and anatase. At pH = 2–3, the prepared  $\text{TiO}_2$  is brookite and anatase. Under these hydrothermal conditions, the  $pK_w$  is about 11.5, and the pH value corresponding to  $h = 3$  is 2.5. However, at pH = 2.5 and 3 ( $h \geq 3$ ), the prepared  $\text{TiO}_2$  contains a small amount of brookite, which is also because of the dynamic change in the hydrolysis ratio during the hydrothermal process. At  $h < 4$ ,  $\text{H}^+$  is constantly generated by the condensation reaction of a  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer during the crystallization process, resulting in a continuous decrease in the pH value, as shown in eqn (17).



When the initial pH value is 2.5 and 3, the final pH value after the hydrothermal crystallization is 1.2 and 1.3, respectively.

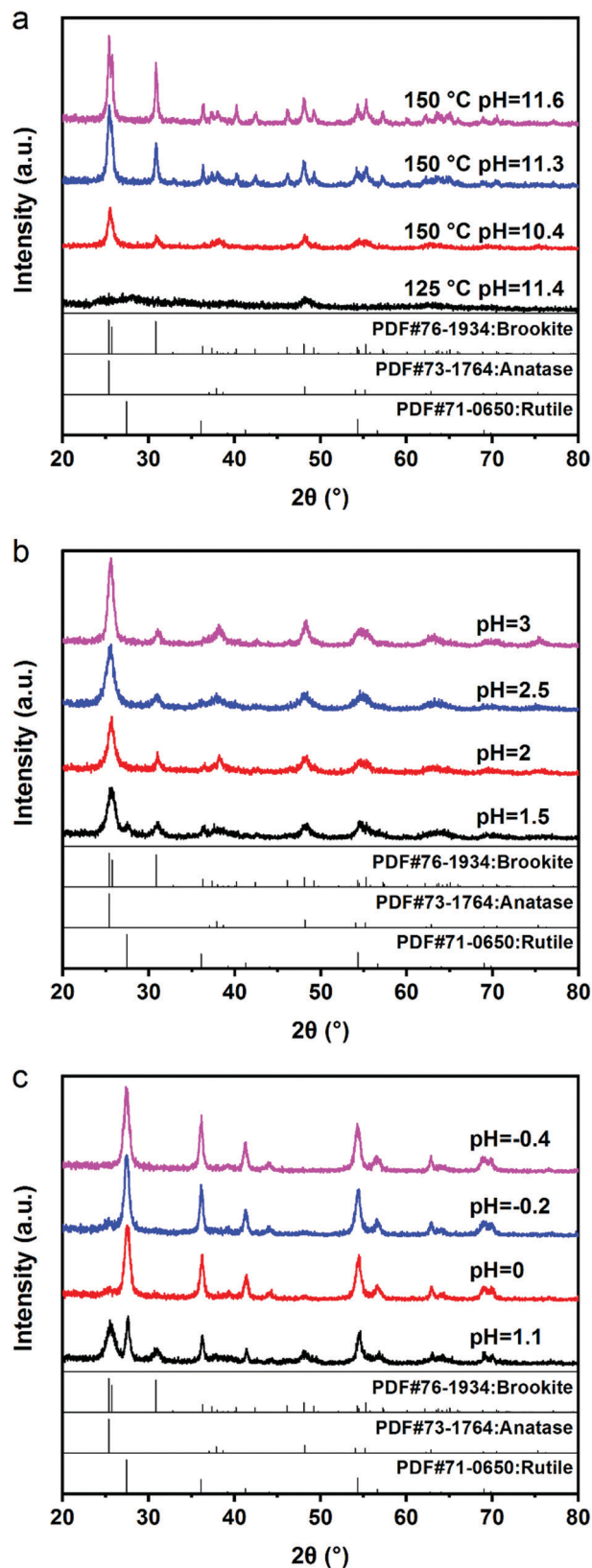


Fig. 8 XRD patterns of the samples synthesized. (a) Around  $h = 5$ ; (b) around  $h = 3$ ; (c) around  $h = 2$ .

Therefore, the hydrolysis ratio continuously decreases during the crystallization process and eventually decreased to the value

of 2.5. As brookite can be obtained at  $2 < h < 3$ , there is a small amount of brookite at pH = 2.5 and 3.

At pH = 0 and −0.02, the prepared  $\text{TiO}_2$  is mainly rutile with a small amount of brookite. At pH = −0.04, the prepared  $\text{TiO}_2$  is purely rutile. Under these hydrothermal conditions, the  $\text{p}K_w$  is about 11.5, and the pH value corresponding to  $h = 2$  is −0.02. However, at pH = −0.02, the prepared  $\text{TiO}_2$  contains a small amount of brookite. This is not caused by the dynamic change in the hydrolysis ratio. According to eqn (17), when  $h = 2$ ,  $\text{H}^+$  is constantly generated and the pH value decreases during the crystallization process, so the hydrolysis ratio is always less than 2 and the brookite structure cannot be obtained. Although the reason is not clear, compared with the experimental result of pH = 1.1, it is obvious that the experimental result corresponding to the correction of the hydrolysis ratio is more consistent with the theoretical result of the crystal formation mechanism.

The experimental results show that the dynamic change in the initial hydrolysis ratio of around 3 and 5 may lead to a mixed crystal structure of  $\text{TiO}_2$ , and it is necessary and reasonable to use the ionic product of water to correct the relationship between the hydrolysis ratio and pH value. The experimental results of different hydrolysis ratios are in agreement with the theoretical results.

## Conclusion

The crystal structure of  $\text{TiO}_2$  is directly affected by the structure of a  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer, and a new crystal formation mechanism of  $\text{TiO}_2$  polymorphs by the condensation of a  $[\text{Ti}(\text{OH})_h(\text{H}_2\text{O})_{6-h}]^{4-h}$  monomer is proposed. The relationship between the hydrolysis ratio and the crystal structure was obtained, and  $h \leq 2$ ,  $2 < h < 3$ ,  $3 \leq h < 5$  and  $h \geq 5$  are the four crystal formation intervals. When  $h \leq 2$ , the structure was crystallized to rutile. When  $2 < h < 3$ , a mixed crystal of rutile, brookite and anatase is seen. In this case, when  $h$  is close to 2, the rutile structure is seen in a higher proportion, whereas when  $h$  is close to 3, the anatase structure is seen in a higher proportion. When  $3 \leq h < 5$ , only the anatase structure will crystallize, whereas when  $h \geq 5$ , only the brookite structure appears. However, there are some deviations between the experimental results of different pH values and the theoretical results corresponding to the hydrolysis ratio calculated at room temperature. The relationship between the hydrolysis ratio and the pH value under hydrothermal conditions is obtained by introducing the ionic product correction and  $h = (3.2507 - 0.1566 \text{p}K_w + 0.3131 \text{pH}) / (0.80168 - 0.00874 \text{p}K_w + 0.01749 \text{pH})$ . The mixed crystal may appear when the initial hydrolysis ratio is between 3 and 5. This is due to the dynamic change in the hydrolysis ratio during the hydrothermal crystallization process. The crystal structure of  $\text{TiO}_2$  synthesized by the hydrothermal method under different hydrolysis ratios is in agreement with the theoretical prediction, which verifies the correctness of the formation mechanism of  $\text{TiO}_2$  polymorphs and the correction method for calculating the hydrolysis ratio. The

crystal structure of  $\text{TiO}_2$  prepared by the hydrothermal synthesis can be easily predicted by our crystal formation mechanism.

## Conflicts of interest

There are no conflicts of interest to declare.

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